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Direct device electrical connection to flexible circuits or other conductors

Abstract

An electrical system comprises a flexible circuit (FC) comprising at least one conductive circuit trace disposed within or between one or more dielectric layers and defining at least one exposed conductive circuit trace portion, an electrical device system comprising an electrical device disposed within a housing and at least one conductive terminal electrically connected to the electrical device and that extend substantially to an end of the housing, and an electrical connector configured to physically secure and electrically connect the at least one exposed conductive circuit trace portion to the at least one conductive terminal of the electrical device system, wherein the electrical connector and the housing of the electrical device system define at least one connection feature for physically securing to each other.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. patent application Ser. No. 17/471,916, filed Sep. 10, 2021, which claims the benefit of U.S. Provisional Application No. 63/077,855, filed Sep. 14, 2020. The entire disclosures of the above applications are incorporated by reference.

FIELD

(1) The present disclosure relates to automotive electrical connectors and, more particularly, to systems and methods for direct device electrical connection to flexible circuits (FCs) or other conductors.

BACKGROUND

(2) A flexible circuit (FC) comprises one or more conductive circuit traces disposed within or between one or more dielectric layers. One conventional method for electrically connecting an FC to another electrical system involves permanently connecting the FC to the other electrical system via soldering, crimping, or similar means. FIGS. 1A-1B, for example, illustrate an example electrical system **100** according to the prior art. An electrical device system **102** comprises an electrical device **104** (e.g., an automotive radar device) housed by a housing **108** to form a male electrical connector that electrically connects to a female electrical connector **110** via a connection position assurance (CPA) feature **128** (e.g., a clip or notch). The female electrical connector **110** comprises a housing **112** that houses one or more round wires **116** with corresponding terminal(s) **124** that extend through the housings **108** and **112** to the electrical device **104**. These round wire(s) **116** are also permanently connected to the conductive circuit trace(s) of an FC **120** (not shown) via soldering, crimping, or similar means. It would be desirable to provide an electrical connection between an FC and another electrical system that is detachable or removable (e.g., when component damage occurs). Additionally, conventional FC electrical connectors are bulky and can take up valuable packaging space and/or increase weight. Thus, while conventional FC connection methods work for their intended purpose, there exists an opportunity for improvement in the relevant art.

(3) The background description provided herein is for the purpose of generally presenting the context of the disclosure. Work of the presently named inventors, to the extent it is described in this background section, as well as aspects of the description that may not otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted as prior art against the present disclosure.

SUMMARY

(4) According to one aspect of the present disclosure, an electrical system, is presented. In one exemplary implementation, the electrical system comprises a flexible circuit (FC) comprising at least one conductive circuit trace disposed within or between one or more dielectric layers and defining at least one exposed conductive circuit trace portion, an electrical device system comprising an electrical device disposed within a housing and at least one conductive terminal electrically connected to the electrical device and that extend substantially to an end of the housing, and an electrical connector configured to physically secure and electrically connect the at least one exposed conductive circuit trace portion to the at least one conductive terminal of the electrical device system, wherein the electrical connector and the housing of the electrical device system define at least one connection feature for physically securing to each other.

(5) In some implementations, the at least one connection feature comprises two opposing clips defined by the housing of the electrical device that are configured to be mated to two corresponding opposing apertures defined by the electrical connector. In some implementations, the at least one connection feature comprises two opposing edges defined by the housing of the electrical device system that are configured to be slidably inserted into two corresponding opposing slots defined by the electrical connector. In some implementations, the FC further defines a stepped configuration when physically secured by the electrical connector. In some implementations, the electrical connector defines a slot that is configured to be engaged by an end effector of a robotic installer to at least one of (i) physically connect the electrical connector to the electrical device system with the FC therebetween and (ii) move and install the electrical system within another electrical system. In some implementations, the electrical connector defines at least one graspable feature that is

configured to be grasped by an end effector of a robotic installer to at least one of (i) physically connect the electrical connector to the electrical device system with the FC therebetween and (ii) move and install the electrical system within another electrical system.

(6) In some implementations, the FC comprises three conductive circuit traces and defines three exposed conductive circuit trace portions, and wherein the electrical device system comprises three conductive terminals. In some implementations, the three conductive terminals are spring terminals that are configured to forcibly contact the three exposed conductive circuit trace portions of the FC to electrically connect the electrical device to the FC. In some implementations, the electrical device is a radio wave detection and ranging (RADAR) device. In some implementations, the at least one conductive terminal comprises a split-blade terminal having two prongs separated by a distance and configured to interface with another electrical terminal of the electrical device, and a spring terminal configured to mate with at least one exposed conductive circuit trace portion of the at least one conductive circuit trace of the FC, wherein the spring terminal and the split-blade terminal are positioned on a longitudinal axis at opposing ends of the electrical connector, and wherein the spring terminal comprises bifurcated contacts that are configured to interface with the at least one exposed conductive circuit trace portion of the at least one conductive circuit trace of the FC.

(7) According to another aspect of the present disclosure, an electrical system is presented. In one exemplary implementation, the electrical system comprises a flexible circuit (FC) comprising a conductive circuit trace disposed within or between one or more dielectric layers and defining one or more connection features formed by laser processing, and a first electrical connector configured to physically secure and electrically connect the FC to a corresponding second electrical connector, wherein the first electrical connector comprises first and second members configured to connect to each other with the FC secured therebetween and to electrically connect to the FC using the one or more connection features.

(8) In some implementations, the FC defines first and second portions having different widths via the laser processing, and wherein a wider of the first and second portions is one or the one or more connection features. In some implementations, the first electrical connector further comprises a seal disposed between the first and second connectable members and surrounding the FC such that it is sealed between the first and second connectable members. In some implementations, the one or more connection features comprise an exposed portion of the conductive circuit trace where at least some of the one or more dielectric layers has been removed via the laser processing. In some implementations, the first and second connectable members define a first set of connection position assurance (CPA) features for establishing the connection therebetween. In some implementations, one of the first and second connectable members comprises a spring terminal configured to contact the exposed portion of the conductive circuit trace to establish the electrical connection therewith, wherein the spring terminal is electrically connectable to the second electrical connector.

(9) In some implementations, the first and second connectable members are each substantially flat such that the FC is substantially perpendicular to a connection direction to the second electrical connector. In some implementations, the first and second connectable members define an angle such that the FC is angled with respect to a connection direction to the second electrical connector. In some implementations, one of the first and second connectable members of the first electrical connector and the second electrical connector define a second set of CPA features for establishing a connection therebetween. In some implementations, one of the first and second connectable members comprises a split-blade terminal having two prongs separated by a distance and configured to interface with another electrical terminal of the second electrical connector, and a spring terminal configured to mate with the exposed portion of the conductive circuit trace of the FC, wherein the spring terminal and the split-blade terminal are positioned on a longitudinal axis at opposing ends of the first electrical connector, and wherein the spring terminal comprises bifurcated contacts that are configured to interface with the exposed portion of the conductive

circuit trace of the FC.

(10) Further areas of applicability of the present disclosure will become apparent from the detailed description provided hereinafter. It should be understood that the detailed description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present disclosure will become more fully understood from the detailed description and the accompanying drawings, wherein:
- (2) FIGS. **1A-1B** illustrate a view of an example electrical system connecting a device to a flexible circuit (FC) according to the prior art;
- (3) FIGS. **2A-2B** illustrate views of a first example electrical system according to some implementations of the present disclosure;
- (4) FIGS. **3A-3E** illustrate views of second and third example electrical systems according to some implementations of the present disclosure;
- (5) FIGS. **4A-4B** illustrate views of example robotic installation features for the example electrical systems according to some implementations of the present disclosure;
- (6) FIG. **5** illustrates a flow diagram of an example method of forming an electrical system according to some implementations of the present disclosure;
- (7) FIG. **6** illustrates a view of another example electrical system connecting a device to an FPC according to the prior art;
- (8) FIGS. **7A-7B** illustrate views of a fourth example electrical system according to some implementations of the present disclosure;
- (9) FIGS. **8A-8B** illustrate views of a fifth example electrical system according to some implementations of the present disclosure;
- (10) FIG. **9** illustrates a view of an alternate configuration of the FC electrical connector of FIGS. **8A-8B**;
- (11) FIG. **10** illustrates a flow diagram of another example method of forming an electrical system according to some implementations of the present disclosure;
- (12) FIG. **11** illustrates a top front perspective view of an example implementation of an example electrical connector according to some implementations of the present disclosure;
- (13) FIG. **12** illustrates an example implementation of the spring terminal from FIG. **11** connecting to the FC according to some implementations of the present disclosure;
- (14) FIG. **13** illustrates a top front perspective view of an example connector in which the electrical connector from FIG. **11** can be used to connect with the FC according to some implementations of the present disclosure; and
- (15) FIG. **14** illustrates a sectional view of the electrical connector in FIG. **11**, taken along section line **14-14**.

DETAILED DESCRIPTION

(16) Referring now to FIGS. **2A-2B**, a first example electrical system **200** according to the principles of the present disclosure is illustrated. While the illustrations shown and described herein are in relation to the same electrical device system **102** of FIG. **1A-1B**, it will be appreciated that the electrical systems of the present disclosure could be applicable to any suitable systems/devices, including, but not limited to, automotive RADAR, LIDAR, lighting devices, and electric motors. The electrical system **200** comprises an electrical connector **204** that is configured to physically secure and electrically connect a flexible circuit (FC) **208** to the electrical device system **102**. While FCs are primarily discussed herein, it will be appreciated that the systems and methods of

the present disclosure are applicable to any suitable conductors, such as any flexible circuits (FCs), of which FCs are a specific type. It will also be appreciated that while the term flexible printed circuit (FPC) is used herein, this is merely one example type or configuration of an FC. The FC **208** comprises three conductive circuit traces **212** disposed within or between one or more dielectric layers **216**. While three conductive circuit traces **216** are shown (e.g., power, ground, and signal), it will be appreciated that the FC **208** could comprise any suitable number of conductive circuit traces **212**. The FC **208** is different from the FC **120** of FIG. **1A-1B** in that it has been pre-processed to define one or more exposed portions (not shown) of the conductive circuit traces **216** on its bottom side. This processing could include, for example, laser processing the FC **208** to remove at least a portion of the dielectric layer **212** to expose the portion(s) of the conductive circuit traces **216**. While laser processing is specifically described herein, it will be appreciated that other processing techniques could be utilized (e.g., physical cutting or stripping).

(17) This exposed portion of the conductive circuit traces **216** are electrically contacted by spring terminals **232** at the ends of terminals (not shown) extending through a portion **228** of the electrical connector **204**. This connection is made when a housing **220** and a cap **224** of the electrical connector **204** are physically secured together. As shown, the housing **220** of the electrical connector **204** is secured to the housing **108** of the electrical device system **102** via a clip or notch feature **236**. The cap **224** of the electrical connector **204** is secured to the housing **220** of the electrical connector **204** via a lip or notch feature **240** of the housing **220** that slidably engages with a corresponding or complimentary groove feature **244** of the cap **224**. It will be appreciated that the FC **208** could also define other connection features, such as a uniquely shaped portion of the FC **208** (e.g., a wider portion) or a unique identifier or barcode (e.g., etched into the dielectric layer **216**). With housings **108** and **220** securely connected via clip or notch feature **236** and the FC **208** securable connected between housing **220** and cap **224** of the electrical connector **204**, the spring terminal(s) **232** forcibly contact the exposed portion(s) of the conductive circuit trace(s) **212** thereby forming a good quality electrical connection between the FC **208** and the electrical device **104** of the electrical device system **102**.

(18) Referring now to FIGS. **3A-3E**, second and third example electrical systems **300**, **350** according to some implementations of the present disclosure are illustrated. While the illustrations shown and described herein are in relation to the same electrical device system **102** of FIG. **1A-1B**, it will be appreciated that the electrical systems of the present disclosure could be applicable to any suitable systems/devices, including, but not limited to, automotive RADAR, LIDAR, lighting devices, and electric motors. In FIG. **3A**, the fifth example electrical system **300** is illustrated. In this configuration, a cap-type electrical connector **304** physically secures to the housing **108** of the electrical device system **102** via a clip or notch feature **308** that is corresponds or is complimentary to an aperture or hole feature **312** in the cap-type electrical connector **304**. The securing force causes exposed portion(s) (not shown) of the conductive circuit trace(s) **212** to electrically contact corresponding terminals (not shown), such as spring terminals, housed within the housing **108** and that electrically connect to the electrical device **104**.

(19) In FIGS. **3B-3C**, the third example electrical system **350** is illustrated. This configuration is similar to but less complex and bulky than the second electrical system **200** of FIGS. **2A-2B**. As shown, an electrical connector **354** (similar to electrical connector **204**) that is physically securable to a shorter housing **358** of the electrical device system **102**. FIG. **3D**, for example, illustrates a cross-sectional view where terminals **378** electrically connected to the electrical device **104** extend substantially to an end **380** of the housing **358**. Spring terminals **368** defined by the ends of the terminals **378** forcibly contact the exposed portion(s) of the conductive circuit trace(s) **212** when the FC **208** is physically secured between the housing **358** and the cap **362** via the corresponding or complimentary lip/notch features **370** of the housing **358** and the groove features **374** of the cap **362**. An optional seal **368** is provided between these members to provide a more insulated electrical connection therein. This configuration may also provide for a stepped configuration or feature **384**

for the FC **208** as shown in FIG. 3E, which could provide to be useful or beneficial for unique packaging situations.

(20) The installation/removal of the FC **208** in the example electrical systems as illustrated and described herein could be performed manually by a human installer or robotically by a robotic installer. FIGS. 4A-4B, for example, illustrate two example robotic installation/removal examples **400A**, **400B**. A robotic installer **404** could be controlled, for example by an electronic controller (not shown) according to a set of computer-executable assembly or installation instructions. Robotic installation, for example, could save costs compared to human installation. In the first example **400A**, a first end effector assembly **408** (e.g., end effector **412** that controls one or more shafts **416**) is controlled by the robotic installer **404** to engage or disengage with robotic installation feature **420**, which is illustrated as a cross-shaped or x-shaped slot. This robotic installation feature **420** can be defined by or attached to cap **428** of an electrical connector **424** (e.g., similar to electrical connector **200**), which further comprises a housing **432** for secure connection to the electrical device system **102**. It will be appreciated that other robotic installation features could be utilized (e.g., scannable identifiers or barcodes). For example, in the other example **400B**, a second end effector assembly **462** (e.g., end effector **466** and a grasping assembly **474**, such as two or four grasping fingers **470**) is controlled by the robotic installer **404** to engage or disengage with robotic installation feature **478** (e.g., graspable edges) of an electrical connector **474**. The electrical connector **474** is physically secured to a housing **458** of a different type of electrical device system **452**, with a similar or different type of electrical device **454** disposed therein.

(21) Referring now to FIG. 5, an example method **500** of forming an electrical system according to the principles of the present disclosure is illustrated. This method **500** could be applicable, for example, to any of the systems described above and illustrated in FIGS. 2A-2B, 3A-3E, and 4A-4B. At **504**, the FC is obtained. At **508**, the FC is processed (e.g., via laser processing or another suitable processing technique, such as manual cutting/stripping) to expose portion(s) of the conductive trace(s) therein. This processing could optionally include processing the FC to define other connection features (different widths, barcodes, etc.). At **512**, the electrical device system is provided with an electrical device disposed within a housing. At **516**, the electrical connector is provided for directly connecting the FC to the electrical device system. At **520**, the electrical connector and the electrical device system are physically secured together with the FC therebetween and electrically connected to the electrical device of the electrical device system. The method **500** then ends or returns to **504** for one or more installation cycles. It will be appreciated that the method **500** could be performed manually by a human assembler/installer, robotically, or in some combination thereof as previously described herein. It will also be appreciated that at least some of these steps of method **500** could be performed in an opposing order to uninstall or disassemble the FC from the electrical device system (e.g., in order to replace a damaged FC).

(22) Another conventional method for electrically connecting an FC to another electrical system involves permanently connecting the FC to the other electrical system via soldering, crimping, or similar means. FIG. 6, for example, illustrates an example electrical system **600** according to the prior art. A device **604** (e.g., an automotive radar device) defines a male electrical connector **608** that electrically connects to a female electrical connector **612**. The female electrical connector **612** includes one or more round wires **616**, which are then permanently connected to the conductive circuit trace(s) of an FC via soldering, crimping, or similar means. It would be desirable to provide an electrical connection between an FC and another electrical system that is detachable or removable (e.g., when component damage occurs). Additionally, conventional FC electrical connectors are bulky and can take up valuable packaging space and/or increase weight. Thus, while conventional FC connection methods work for their intended purpose, there exists an opportunity for improvement in the relevant art.

(23) Referring now to FIGS. 7A-7B, a fourth example electrical system **700** according to the principles of the present disclosure is illustrated. While the illustrations shown and described herein

are in relation to the same device **604** of FIG. **6**, it will be appreciated that the electrical systems of the present disclosure could be applicable to any suitable devices, including, but not limited to, automotive RADAR, LIDAR, lighting devices, and electric motors. The electrical system **700** comprises a first electrical connector **704** that is configured to physically secure and electrically connect a flexible FC **708** to the second electrical connector **608** of the device **604**. The FC **708** comprises three conductive circuit traces **712** disposed within or between one or more dielectric layers **716**. While three conductive circuit traces **716** are shown (e.g., power, ground, and signal), it will be appreciated that the FC **708** could comprise any suitable number of conductive circuit traces **712**. The FC **708** is different from the FC **620** of FIG. **6** in that it has been pre-processed to define one or more connection features **720**. This processing could include, for example, laser processing the FC **708** to alter its shape and/or to remove at least a portion of the dielectric layer **712** to expose a portion of the conductive circuit traces **716**. While laser processing is specifically described herein, it will be appreciated that other processing techniques could be utilized (e.g., physical cutting or stripping).

(24) As shown, a portion of the dielectric layer **716** has been removed via laser processing such that a portion of the conductive circuit traces **712** are exposed. This exposed portion of the conductive circuit traces **716** is one of the one or more connection features **720** (e.g., exposed portions of the conductive circuit traces **712**). It will be appreciated that the FC **708** could define other connection features, such as a uniquely shaped portion (described in greater detail later herein) or a unique identifier or barcode (e.g., etched into the dielectric layer **716**). The fourth electrical connector **704** comprises first and second connectable members **724**, **728**. While the first connectable member **724** is illustrated as a connector (male) that is inserted into the second electrical connector **608** (female), it will be appreciated that the opposite configuration could be utilized. The first and second connectable members **724**, **728** are connectable using a first set of connection position assurance (CPA) features **732**. While a snap-in type CPA feature **732** is illustrated, it will be appreciated that other suitable CPA feature configurations could be utilized. The first connectable member **724** is similarly connectable to the second electrical connector **608** using a second set of CPA features **736**. Once the FC **708** is properly aligned (e.g., using the connection feature(s) **720**), the first and second members **724**, **728** are connected thereby physically securing the FC **708** therein. A seal **740** could be optionally included to seal and protect the portion of the FC **708** from the environment (dust, moisture, etc.).

(25) The first connectable member **724** also houses electrical connections **744**, which engage the exposed portions of the conductive traces **712**. These electrical connections **744** could be spring-loaded connections that slightly compress when the FC **708** is forced down thereupon in response to the first and second connectable members **724**, **728** being connected. It will be appreciated that any suitable type or configuration of the electrical connections **744** could be utilized. These electrical connections **744** in turn connect to respective electrical connections (not shown) of the second electrical connector **608**. The installation/removal of the FC **708** within/from the first electrical connector **704** and the installation/removal of the first electrical connector **704** (with the FC **708** therein) with/from the second electrical connector **608** could be performed manually by a human installer or robotically by a robotic installer **624** in any suitable combination. The robotic installer **624** could be controlled, for example by an electronic controller (not shown) according to a set of computer-executable assembly or installation instructions. Robotic installation, for example, could save costs compared to human installation. In the robotic installation example, an end effector **628** of the robotic installer **624** could interact with robotic installation features (identifiers, engageable slots, etc.). A specific robotic installation feature will be shown in FIGS. **8A-8B** and FIG. **9** and described in greater detail below.

(26) Referring now to FIGS. **8A-8B**, a fifth example electrical system **800** according to the principles of the present disclosure is illustrated. Again, while the illustrations shown and described herein are in relation to the same device **604** of FIG. **6**, it will be appreciated that the electrical

systems of the present disclosure could be applicable to any suitable devices, including, but not limited to, automotive RADAR, LIDAR, lighting devices, and electric motors. The electrical system **800** comprises a first electrical connector **804** that is configured to physically secure and electrically connect an FC **808** to the second electrical connector **608** of the device **604**. The FC **808** comprises two conductive circuit traces **812** disposed within or between one or more dielectric layers **816**. While two conductive circuit traces **816** are shown, it will be appreciated that the FC **808** could comprise any suitable number of conductive circuit traces **812**. The FC **808** is different from the FC **620** of FIG. 6 and the FC **208** of FIG. 2A-2B in that it has been pre-processed to define connection features **820a**, **820b** (e.g., using laser processing and/or other suitable processing techniques). As shown, a portion of the dielectric layer **816** has been removed via laser processing such that a portion of the conductive circuit traces **812** are exposed. This exposed portion of the conductive circuit traces **816** corresponds to a first connection feature **820a**. Also shown is a wider portion of the FC **808** compared to a remainder of the FC **808**. This wider portion of the FC **808** corresponds to a second connection feature **820b**. It will be appreciated that the FC **808** could also include other connection features, such as a unique identifier or barcode (e.g., etched into the dielectric layer **816**).

(27) The first electrical connector **804** comprises first and second connectable members **824**, **828**. While the first connectable member **824** is illustrated as a connector (male) that is inserted into the second electrical connector **608** (female), it will be appreciated that the opposite configuration could be utilized. The first and second connectable members **824**, **828** are connectable using a first set of CPA features **832**. While a clip-type CPA feature **832** is illustrated, it will be appreciated that other suitable CPA feature configurations could be utilized. The first connectable member **824** is similarly connectable to the second electrical connector **608** using a second set of CPA features **836**. Once the FC **808** is properly aligned (e.g., using the connection feature(s) **820a**, **820b**), the first and second members **824**, **828** are connected thereby physically securing the FC **808** therein. A seal **840** could be optionally included to seal and protect the portion of the FC **808** from the environment (dust, moisture, etc.). The first electrical connector **804** may be superior to the fifth electrical connector **704** of FIGS. 7A-7B in that the FC **808** has been processed to define the second connection feature **820b**, which could further assist in the alignment/installation of the FC **808** and/or could provide for improved securability of the FC **808** therein.

(28) As previously discussed, the installation or assembly could be performed manually by a human installer or robotically by a robotic installer **624** in any suitable combination. In the robotic installation example, an end effector **628** of the robotic installer **624** could interact with robotic installation feature **848**, which is illustrated as a hexagonal-shaped slot. It will be appreciated that other robotic installation features could be utilized (e.g., scannable identifiers or barcodes). In the illustrated example, the end effector **628** of the robotic installer **624** is configured to interact with the robotic installation feature **848**. For example only, the end effector **628** could comprise two or more members each defining an L-shape such that their distal ends can engage a base or bottom portion of the robotic installation feature **848**. It will be appreciated that other shaped or configured robotic installation features and/or end effectors could be utilized. Non-limiting examples of the robotic installation feature **848** include a gripping or grasping member (e.g., a clamp) that is configured to grip or grasp an X-shaped gripping or grasping feature that is attached to the outer surface of the first electrical connector **804**. Again, it will be appreciated that other shapes could be utilized (a round bar, a flat bar, a circular disc, an equal armed cross shape, a t-shape, a keyhole slot, etc.).

(29) Referring now to FIG. 9, an alternate configuration **900** of the first electrical connector **804** of the electrical system **800** of FIGS. 8A-8B (hereinafter, "first electrical connector **900**") according to the principles of the present disclosure is illustrated. While not shown, the first electrical connector **900** is similarly connectable to the second electrical connector **608** of device **604** as illustrated in FIGS. 8A-8B. The first electrical connector **900** provides a different configuration than the first

electrical connector **804** of FIGS. **8A-8B**. More particularly, in the electrical system **800** of FIGS. **8A-8B**, the first and second connectable members **824**, **828** of the first electrical connector **804** are each substantially flat such that the FC **808** is substantially perpendicular to a connection direction **852** (i.e., vertical) to the second electrical connector **608**. In contrast, in the first electrical connector **900** of FIG. **9**, the first and second connectable members **920**, **924** define an angle **948** such that the FC **904** is angled with respect to the connection direction **952** (i.e., vertical) to the second electrical connector (i.e., second electrical connector **608**). The first electrical connector **900** therefore may be desirable for specific installations, such as an installation having ample vertical packaging space but limited lateral packaging space. The remaining features **904-944** generally correspond to the same features **804-844** of electrical system **800**, although some are slightly modified due to the angled configuration (e.g., the spring contact electrical connectors **940**).

(30) Referring now to FIG. **10**, another example method **1000** of forming an electrical system according to the principles of the present disclosure is illustrated. This method **1000** could be applicable, for example, to any of the systems described above and illustrated in FIGS. **2A-4B**, **7A-8B**, and **9**. At **1004**, the FC is obtained. At **1008**, the FC is processed (e.g., via laser processing or another suitable processing technique, such as manual cutting/stripping) to define the one or more connection features (exposed conductive circuit traces, a wider shape/profile, etc.). At **1012**, the first electrical connector is obtained for securing and electrically connecting the FC to a second electrical connector (e.g., second electrical connector **108** of device **104**). At **1016**, the FC is securely installed within the first electrical connector (e.g., including electrically connecting the first electrical connector therewith). At **1020**, the first electrical connector is electrically connected to the second electrical connector, thereby electrically connecting the FC to the second electrical connector. The method **1000** then ends or returns to **1004** for one or more installation cycles. It will be appreciated that the method **1000** could be performed manually by a human assembler/installer, robotically, or in some combination thereof as previously described herein. It will also be appreciated that at least some of these steps of method **1000** could be performed in an opposing order to uninstall or disassemble the FC from the first and second electrical connectors (e.g., in order to replace a damaged FC).

(31) FIG. **11** illustrates a top front perspective view **1100** of another example

(32) implementation of an electrical connector terminal. In this illustrated example, the terminal **1100** includes the split-blade terminal **1102** at one end, which has prongs **1106** configured to mate with a blade terminal or a pin terminal. At the opposing end, the electrical connector includes the spring terminal **1104**. The spring terminal **1104** in this example has an arcuate shape (e.g., curved L-shape) and is configured to interface with an FC that does not run perpendicular to a longitudinal axis **1112** of the terminal **1100**. Rather, the spring terminal **1104** in this example has a contact surface **1108** that is within a range of 30 to 60 degrees from the longitudinal axis **1112**. In aspects, the terminal **1100** includes one or more bends (e.g., bend **1106**) such that the spring terminal **1104** and the split-blade terminal **1102** are rotated relative to one another about the longitudinal axis **1112** by approximately 90 degrees. The spring terminal **1104** also includes a protrusion, such as protrusion **1110**.

(33) FIG. **12** illustrates example implementations **1200** of the spring terminal **1104** from FIG. **11** connecting to an FC. View **1200-1** illustrates the spring terminal **1104** approaching the flat-wire conductors **1208** of the wiring **1206** in a direction corresponding to the longitudinal axis **1204** of the terminal **1100**. In view **1200-2**, the terminal **1100** is pressed against the flat-wire conductors **1208**, based on a longitudinal compression force along the longitudinal axis **1204** of the terminal **1100**. Due to the angle of the contact not being perpendicular to the compression force, the protrusion **1110** may slidably move along the flat-wire conductor **1208** a short distance. This slidable movement may wipe away debris that might be on the flat-wire conductor **1208**, providing a clean surface to interface with the spring terminal **1104**.

(34) FIG. **13** illustrates a top front perspective view **1300** of an example connector in which the

spring terminal from FIG. 11 can be used to connect with the FC. Here, the housing 1302 is connected to a retainer 1304 on opposing sides of the wiring 1206. The retainer 1304 compresses the wiring 1206 against the spring terminals 1104 of the terminal 1100 housed within the housing 1302 and secures the wiring 1206 in place, which maintains contact with the terminal 1100.

(35) Lastly, FIG. 14 illustrates a sectional view 1400 of the connector in FIG. 13, taken along section line 14-14. In the sectional view 1400, the spring terminal 204 is contacting the wiring 106 based on a compression force provided by the housing 1302 and the retainer 1304 being fastened together on opposing sides of the wiring 1206. The compression force is in a longitudinal direction of the terminal 1100. The housing 1302 includes an opening to allow an electrical component to mate with the split-blade terminal 202 of the terminal 1100.

(36) Example embodiments are provided so that this disclosure will be thorough, and will fully convey the scope to those who are skilled in the art. Numerous specific details are set forth such as examples of specific components, devices, and methods, to provide a thorough understanding of embodiments of the present disclosure. It will be apparent to those skilled in the art that specific details need not be employed, that example embodiments may be embodied in many different forms and that neither should be construed to limit the scope of the disclosure. In some example embodiments, well-known procedures, well-known device structures, and well-known technologies are not described in detail.

(37) The terminology used herein is for the purpose of describing particular example embodiments only and is not intended to be limiting. As used herein, the singular forms “a,” “an,” and “the” may be intended to include the plural forms as well, unless the context clearly indicates otherwise. The term “and/or” includes any and all combinations of one or more of the associated listed items. The terms “comprises,” “comprising,” “including,” and “having,” are inclusive and therefore specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. The method steps, processes, and operations described herein are not to be construed as necessarily requiring their performance in the particular order discussed or illustrated, unless specifically identified as an order of performance. It is also to be understood that additional or alternative steps may be employed.

(38) Although the terms first, second, third, etc. may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish one element, component, region, layer or section from another region, layer or section. Terms such as “first,” “second,” and other numerical terms when used herein do not imply a sequence or order unless clearly indicated by the context. Thus, a first element, component, region, layer or section discussed below could be termed a second element, component, region, layer or section without departing from the teachings of the example embodiments.

(39) As used herein, the term module may refer to, be part of, or include: an Application Specific Integrated Circuit (ASIC); an electronic circuit; a combinational logic circuit; a field programmable gate array (FPGA); a processor or a distributed network of processors (shared, dedicated, or grouped) and storage in networked clusters or datacenters that executes code or a process; other suitable components that provide the described functionality; or a combination of some or all of the above, such as in a system-on-chip. The term module may also include memory (shared, dedicated, or grouped) that stores code executed by the one or more processors.

(40) The term code, as used above, may include software, firmware, byte-code and/or microcode, and may refer to programs, routines, functions, classes, and/or objects. The term shared, as used above, means that some or all code from multiple modules may be executed using a single (shared) processor. In addition, some or all code from multiple modules may be stored by a single (shared) memory. The term group, as used above, means that some or all code from a single module may be executed using a group of processors. In addition, some or all code from a single module may be

stored using a group of memories.

(41) The techniques described herein may be implemented by one or more computer programs executed by one or more processors. The computer programs include processor-executable instructions that are stored on a non-transitory tangible computer readable medium. The computer programs may also include stored data. Non-limiting examples of the non-transitory tangible computer readable medium are nonvolatile memory, magnetic storage, and optical storage.

(42) Some portions of the above description present the techniques described herein in terms of algorithms and symbolic representations of operations on information. These algorithmic descriptions and representations are the means used by those skilled in the data processing arts to most effectively convey the substance of their work to others skilled in the art. These operations, while described functionally or logically, are understood to be implemented by computer programs. Furthermore, it has also proven convenient at times to refer to these arrangements of operations as modules or by functional names, without loss of generality.

(43) Unless specifically stated otherwise as apparent from the above discussion, it is appreciated that throughout the description, discussions utilizing terms such as “processing” or “computing” or “calculating” or “determining” or “displaying” or the like, refer to the action and processes of a computer system, or similar electronic computing device, that manipulates and transforms data represented as physical (electronic) quantities within the computer system memories or registers or other such information storage, transmission or display devices.

(44) Certain aspects of the described techniques include process steps and instructions described herein in the form of an algorithm. It should be noted that the described process steps and instructions could be embodied in software, firmware or hardware, and when embodied in software, could be downloaded to reside on and be operated from different platforms used by real time network operating systems.

(45) The present disclosure also relates to an apparatus for performing the operations herein. This apparatus may be specially constructed for the required purposes, or it may comprise a general-purpose computer selectively activated or reconfigured by a computer program stored on a computer readable medium that can be accessed by the computer. Such a computer program may be stored in a tangible computer readable storage medium, such as, but is not limited to, any type of disk including floppy disks, optical disks, CD-ROMs, magnetic-optical disks, read-only memories (ROMs), random access memories (RAMs), EPROMs, EEPROMs, magnetic or optical cards, application specific integrated circuits (ASICs), or any type of media suitable for storing electronic instructions, and each coupled to a computer system bus. Furthermore, the computers referred to in the specification may include a single processor or may be architectures employing multiple processor designs for increased computing capability.

(46) The algorithms and operations presented herein are not inherently related to any particular computer or other apparatus. Various general-purpose systems may also be used with programs in accordance with the teachings herein, or it may prove convenient to construct more specialized apparatuses to perform the required method steps. The required structure for a variety of these systems will be apparent to those of skill in the art, along with equivalent variations. In addition, the present disclosure is not described with reference to any particular programming language. It is appreciated that a variety of programming languages may be used to implement the teachings of the present disclosure as described herein, and any references to specific languages are provided for disclosure of enablement and best mode of the present invention.

(47) The foregoing description of the embodiments has been provided for purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure. Individual elements or features of a particular embodiment are generally not limited to that particular embodiment, but, where applicable, are interchangeable and can be used in a selected embodiment, even if not specifically shown or described. The same may also be varied in many ways. Such variations are

not to be regarded as a departure from the disclosure, and all such modifications are intended to be included within the scope of the disclosure.

Claims

1. An electrical system, comprising: a flexible circuit (FC) comprising a conductive circuit trace disposed within or between one or more dielectric layers and defining one or more connection features; and a first electrical connector configured to physically secure and electrically connect the FC to a corresponding second electrical connector, the first electrical connector comprising first and second members configured to connect to each other with the FC secured therebetween and to electrically connect to the FC using the one or more connection features, the first member defining a first side and a second side opposite the first side, the FC extending through the first side and the second side, the FC extending between the first member and the second member in a direction substantially perpendicular to a connection direction to the second electrical connector, the first member defining a slot that is configured to be engaged by an end effector of a robotic installer to at least one of (i) physically connect the first electrical connector to the corresponding second electrical connector with the FC therebetween or (ii) move and install the electrical system within another electrical system.
2. The electrical system of claim 1, wherein the FC defines first and second portions having different widths via laser processing, and wherein a wider of the first and second portions is one of the one or more connection features.
3. The electrical system of claim 1, wherein the first electrical connector further comprises a seal disposed between the first and second members and surrounding the one or more connection features.
4. The electrical system of claim 1, wherein the one or more connection features comprise an exposed portion of the conductive circuit trace.
5. The electrical system of claim 1, wherein the first and second members define a first set of connection position assurance (CPA) features for establishing a connection therebetween.
6. The electrical system of claim 1, wherein the first electrical connector defines a CPA feature for establishing a connection with the second electrical connector.
7. The electrical system of claim 1, further comprising a spring terminal configured to contact a connection feature of the one or more connection features to establish the electrical connection therewith.
8. The electrical system of claim 1, wherein the first and second members are each substantially flat.
9. The electrical system of claim 1, further comprising a first electrical terminal having a first end and a second end, the first end configured to interface with a second electrical terminal of the second electrical connector, the second end configured to interface with a connection feature of the one or more connection features.
10. The electrical system of claim 1, wherein the one or more connection features are formed by laser processing.
11. The electrical system of claim 1, wherein the first member is configured to engage the second member and sealably-fix the FC therebetween.
12. The electrical system of claim 1, wherein the first member defines first and second openings, and wherein the FC extends through the first opening and the second opening.
13. The electrical system of claim 1, wherein the slot is a hexagonal-shaped slot.
14. The electrical system of claim 1, further comprising a seal disposed between the first and second members.
15. The electrical system of claim 14, wherein the seal surrounds the one or more connection features.

16. The electrical system of claim 14, wherein the first member defines an outer edge, and wherein the seal is disposed proximate to the outer edge.
17. The electrical system of claim 1, further comprising: the second electrical connector; and a housing coupled to the second electrical connector.
18. The electrical system of claim 17, further comprising an electrical device disposed within the housing.
19. The electrical system of claim 1, further comprising the second electrical connector, wherein the first electrical connector includes a male connector configuration, and wherein the second electrical connector includes a female connector configuration configured to mate with the male connector configuration.
20. An electrical system, comprising: a flexible circuit (FC) comprising a conductive circuit trace disposed within or between one or more dielectric layers and defining one or more connection features, the FC defining first and second portions having different widths via laser processing, a wider of the first and second portions being one of the one or more connection features; and a first electrical connector configured to physically secure and electrically connect the FC to a corresponding second electrical connector, the first electrical connector comprising first and second members configured to connect to each other with the FC secured therebetween and to electrically connect to the FC using the one or more connection features, the first member defining a first side and a second side opposite the first side, the FC extending through the first side and the second side, the first member defining a slot that is configured to be engaged by an end effector of a robotic installer to at least one of (i) physically connect the first electrical connector to the corresponding second electrical connector with the FC therebetween or (ii) move and install the electrical system within another electrical system.
21. An electrical system, comprising: a flexible circuit (FC) comprising a conductive circuit trace disposed within or between one or more dielectric layers and defining one or more connection features; and a first electrical connector configured to physically secure and electrically connect the FC to a corresponding second electrical connector, the first electrical connector comprising first and second members configured to connect to each other with the FC secured therebetween and to electrically connect to the FC using the one or more connection features, the first member defining a first side and a second side opposite the first side, the FC extending through the first side and the second side, the first member defining a slot that is configured to be engaged by an end effector of a robotic installer to at least one of (i) physically connect the first electrical connector to the corresponding second electrical connector with the FC therebetween or (ii) move and install the electrical system within another electrical system, the first electrical connector including a seal disposed between the first and second members and surrounding the one or more connection features.
22. An electrical system, comprising: a flexible circuit (FC) comprising a conductive circuit trace disposed within or between one or more dielectric layers and defining one or more connection features; and a first electrical connector configured to physically secure and electrically connect the FC to a corresponding second electrical connector, the first electrical connector comprising first and second members configured to connect to each other with the FC secured therebetween and to electrically connect to the FC using the one or more connection features, the first member defining a first side and a second side opposite the first side, the FC extending through the first side and the second side, the first member configured to engage the second member and sealably-fix the FC therebetween, the first member defining a slot that is configured to be engaged by an end effector of a robotic installer to at least one of (i) physically connect the first electrical connector to the corresponding second electrical connector with the FC therebetween or (ii) move and install the electrical system within another electrical system.
23. An electrical system, comprising: a flexible circuit (FC) comprising a conductive circuit trace disposed within or between one or more dielectric layers and defining one or more connection

features; and a first electrical connector configured to physically secure and electrically connect the FC to a corresponding second electrical connector, the first electrical connector comprising first and second members configured to connect to each other with the FC secured therebetween and to electrically connect to the FC using the one or more connection features, the first member defining a first side and a second side opposite the first side, the FC extending through the first side and the second side, the first member defining a hexagonal-shaped slot that is configured to be engaged by an end effector of a robotic installer to at least one of (i) physically connect the first electrical connector to the corresponding second electrical connector with the FC therebetween or (ii) move and install the electrical system within another electrical system.

24. An electrical system, comprising: a flexible circuit (FC) comprising a conductive circuit trace disposed within or between one or more dielectric layers and defining one or more connection features; a first electrical connector configured to physically secure and electrically connect the FC to a corresponding second electrical connector, the first electrical connector comprising first and second members configured to connect to each other with the FC secured therebetween and to electrically connect to the FC using the one or more connection features, the first member defining a first side and a second side opposite the first side, the FC extending through the first side and the second side, the first member defining a slot that is configured to be engaged by an end effector of a robotic installer to at least one of (i) physically connect the first electrical connector to the corresponding second electrical connector with the FC therebetween or (ii) move and install the electrical system within another electrical system; and a seal disposed between the first and second members.

25. The electrical system of claim 24, wherein the first and second members define an angle such that the FC extends between the first and second members in a direction defining an angle relative to a connection direction to the second electrical connector, and wherein the angle is greater than zero degrees and less than ninety degrees.
